

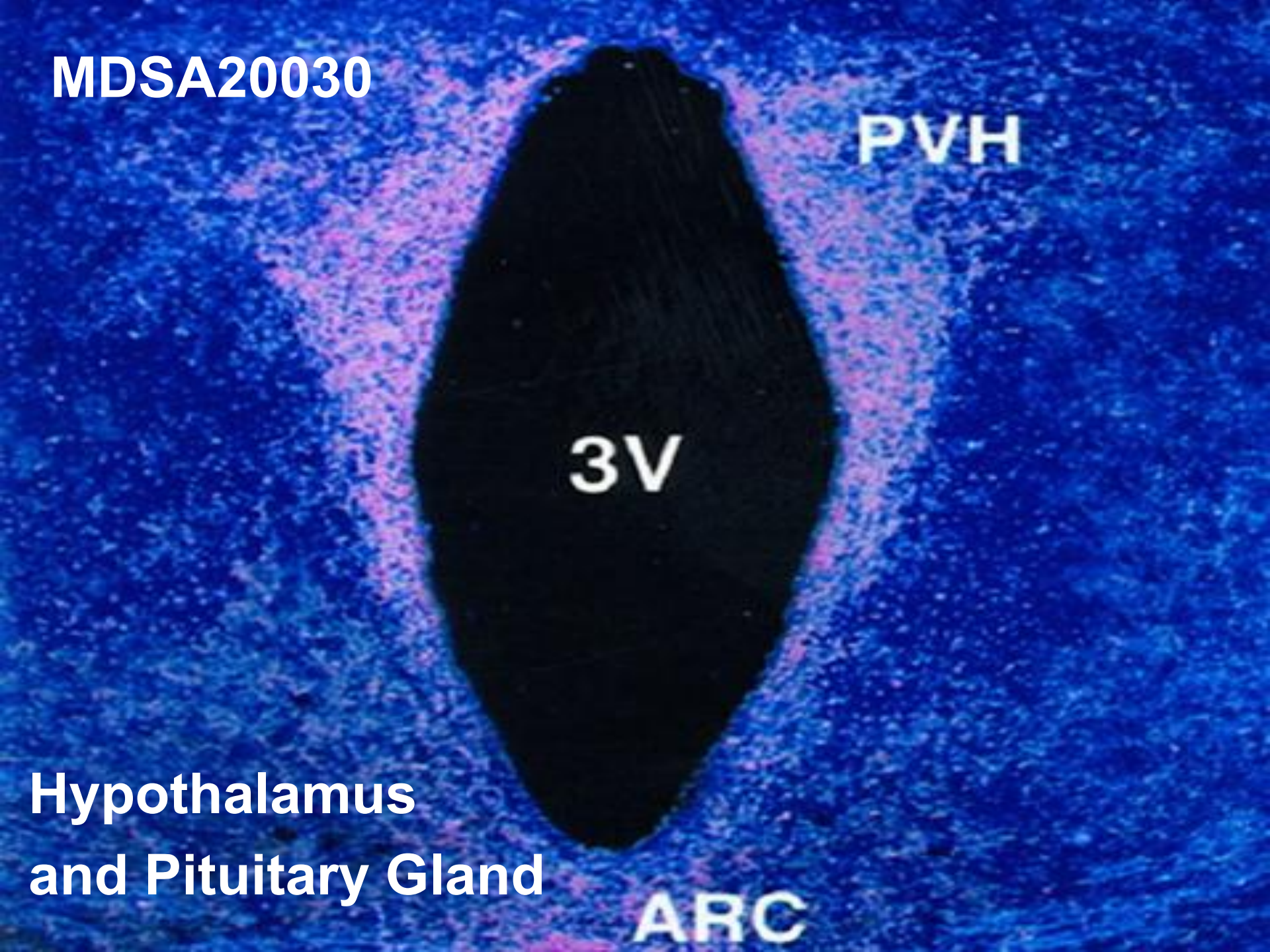
MDSA20030

PVH

3V

**Hypothalamus
and Pituitary Gland**

ARC

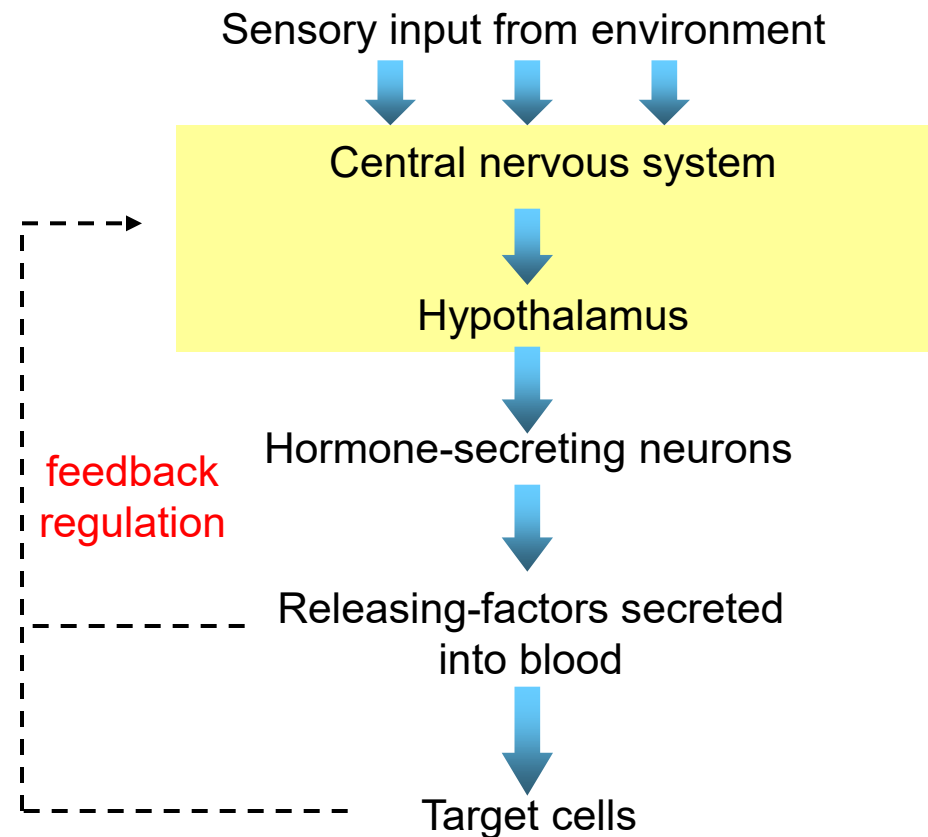


Learning Objectives

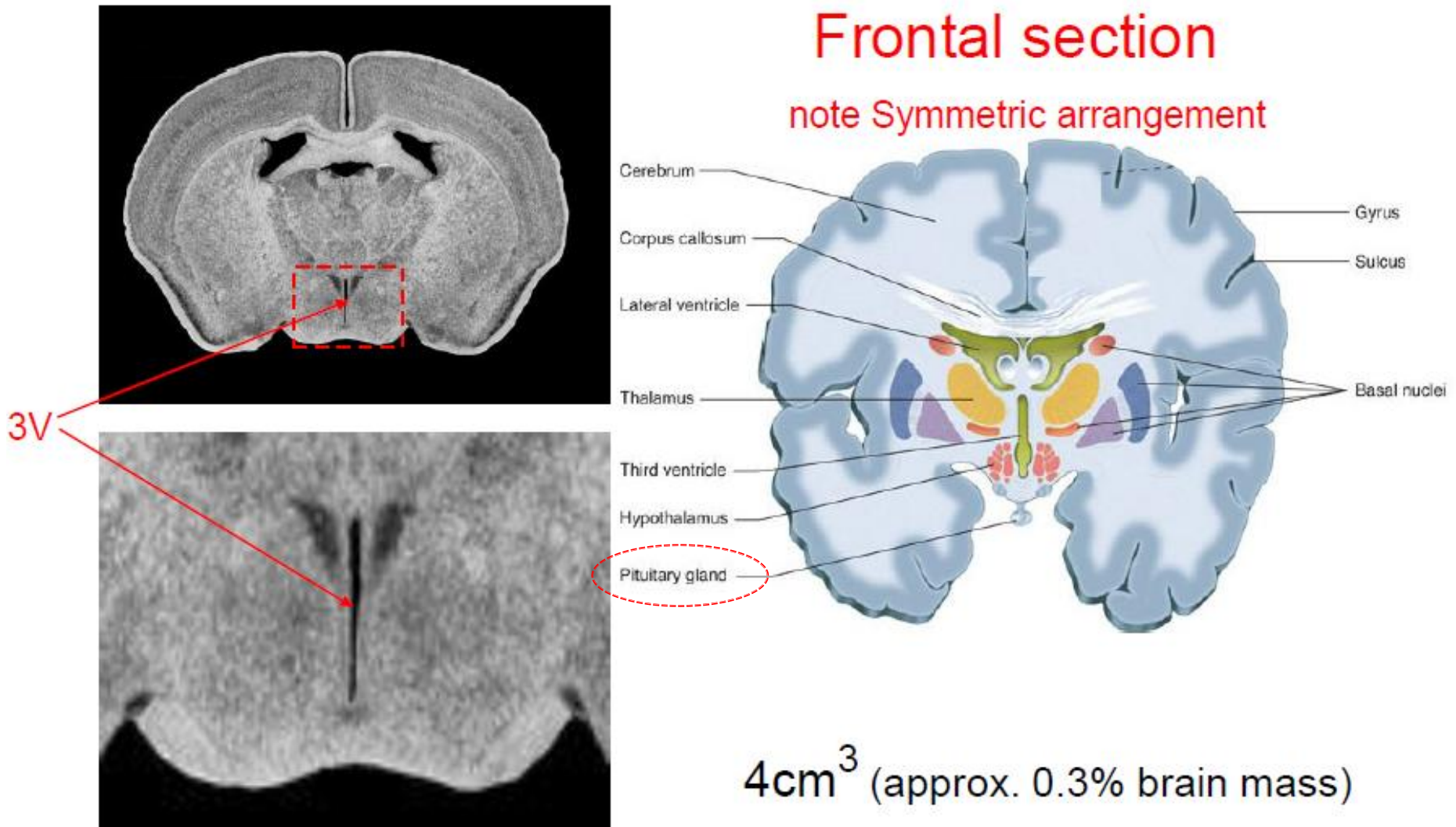
- Understand the structure of the hypothalamus and the concept of hypothalamic nuclei
- Understand the concept of hypothalamic neuroendocrine cells and how they are connected to the pituitary gland
- Know the hypophysiotrophic hormones and their function
- Understand the function of the posterior pituitary hormones vasopressin and oxytocin

Hypothalamus

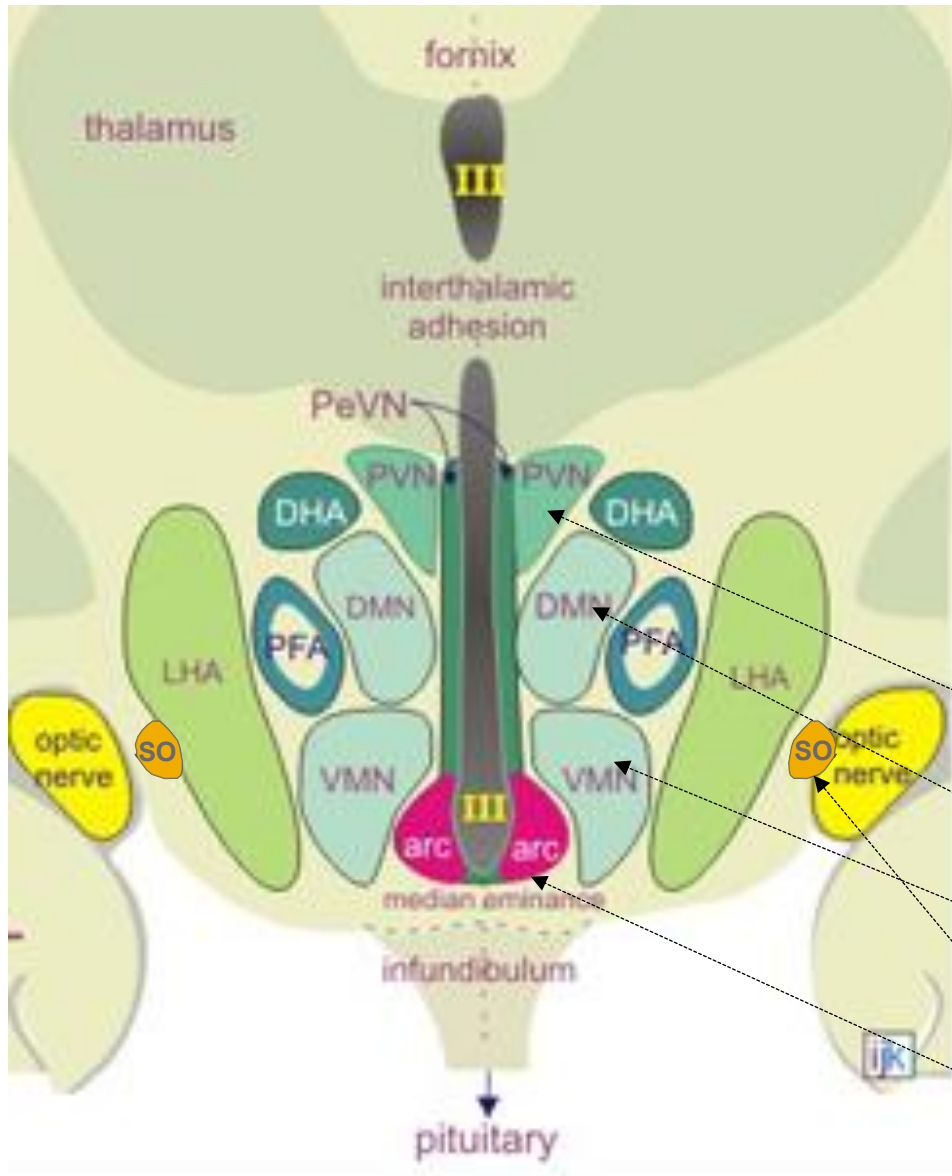
- It is the interface between the nervous and endocrine systems
- Integrates signals from CNS, PNS and endocrine system
- Co-ordinated response
 - visceral ANS (neural output)
 - **neuroendocrine function (pituitary)**
- Main endocrine functions
 - growth
 - fertility and reproduction
 - osmoregulation (water balance)
 - metabolism (thermoregulation)
 - energy balance
 - stress response
 - lactation



Hypothalamus is Located Adjacent to the 3rd Ventricle



Coronal Section Through Hypothalamus



Nuclei - clusters of functionally related neuronal cell bodies. The boundaries between nuclei can be indistinct. Even within nuclei, there may be discrete populations of neuronal populations with different functions

Some nuclei contain specialised neurons, neuroendocrine cells which are connected to the pituitary gland

They release hormones that regulate pituitary function

Paraventricular nucleus

Dorsomedial nucleus

Ventromedial nucleus

Supraoptic nucleus

Arcuate nucleus

Note – peri (PeVN) and paraventricular (PVN) nuclei are distinct

Hypothalamic Neuroendocrine Cells

- Neuroendocrine cells
neurons that release hormones
(into the circulation)

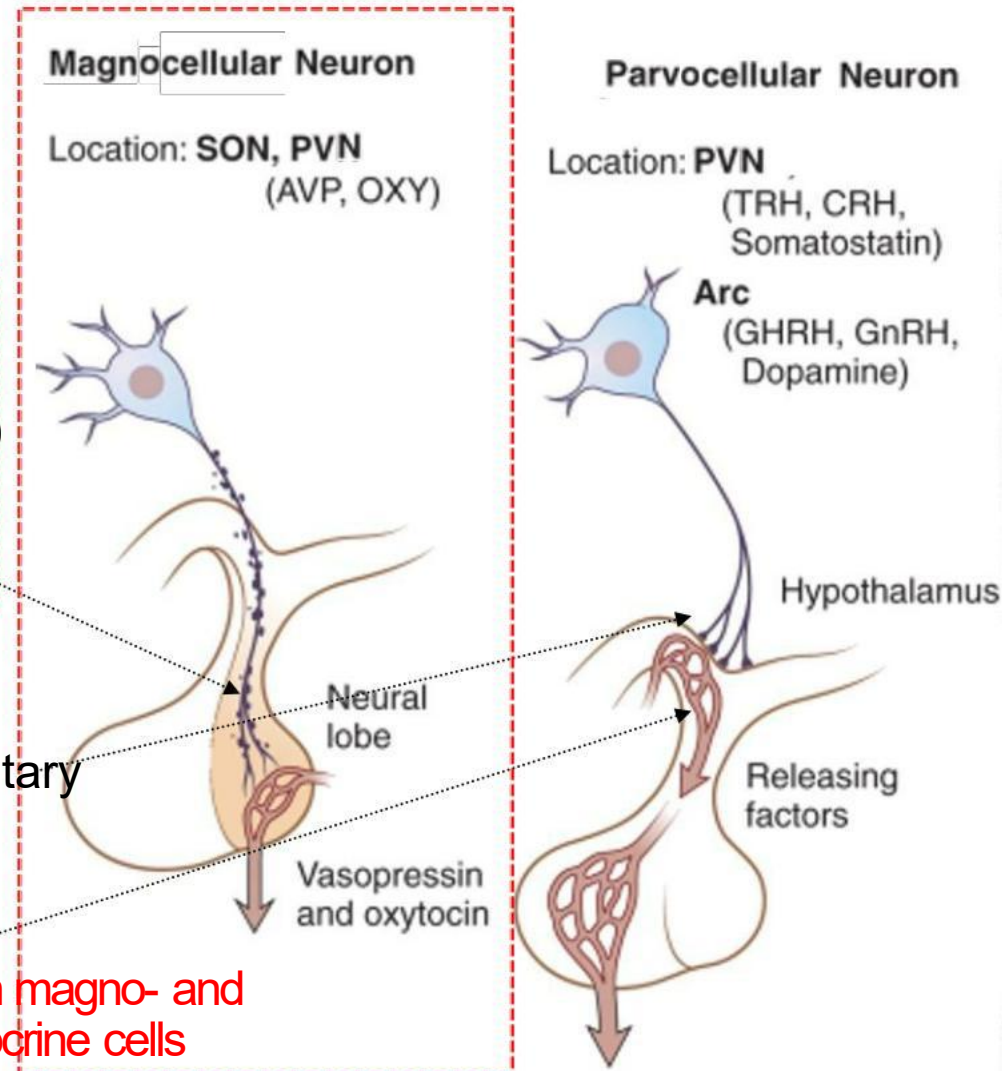
- Magnocellular
large cell bodies, long axons that
project to pituitary (posterior lobe)

Oxytocin and vasopressin

- Parvocellular
Small cell bodies, short axons that
project to infundibulum (top of pituitary
stalk)

TRH, CRH, SST, GHRH, GnRH,
dopamine

Some nuclei have both magno- and
parvocellular neuroendocrine cells



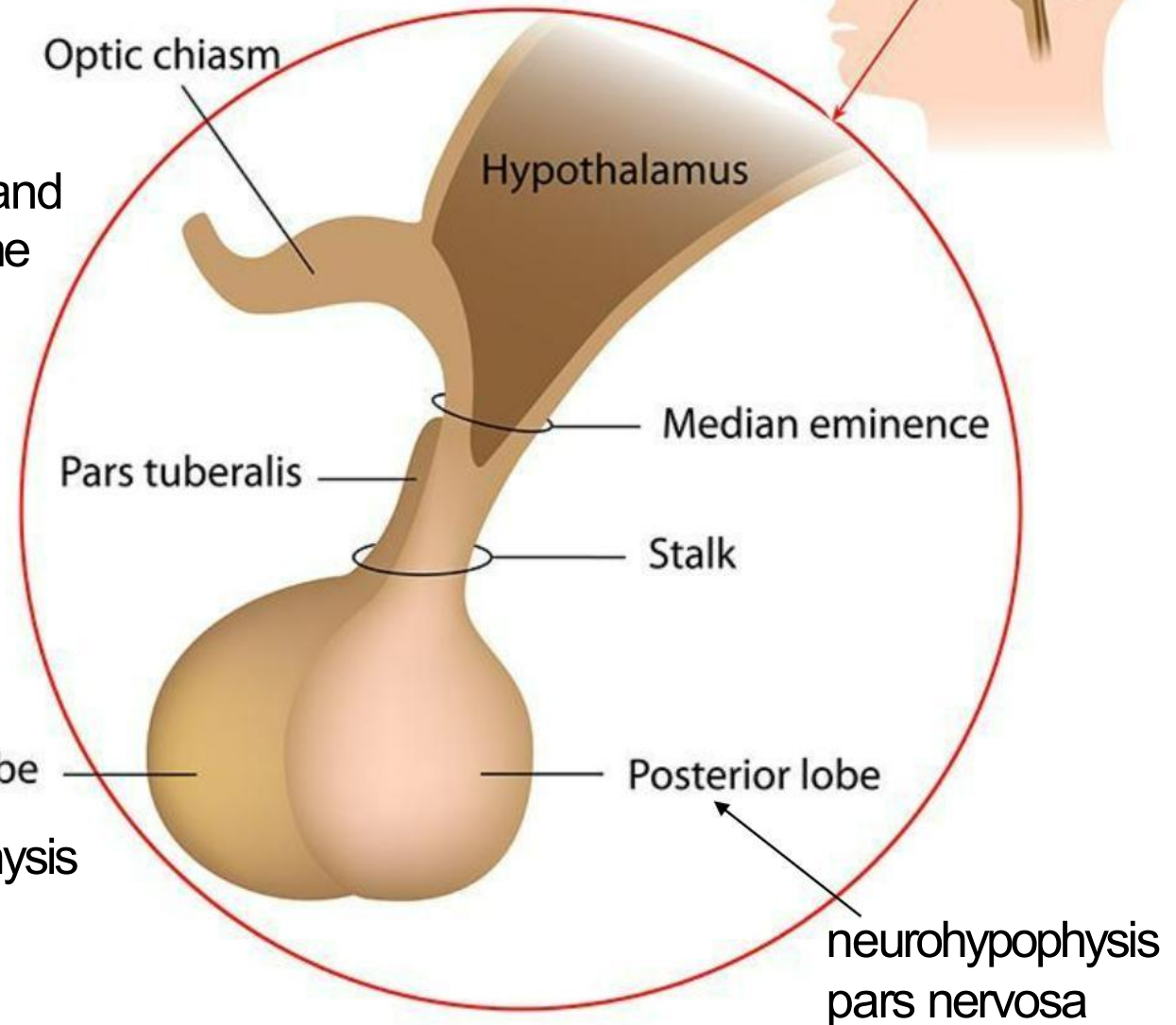
The Pituitary Gland (aka hypophysis)

The pituitary is the **master gland** of the endocrine system

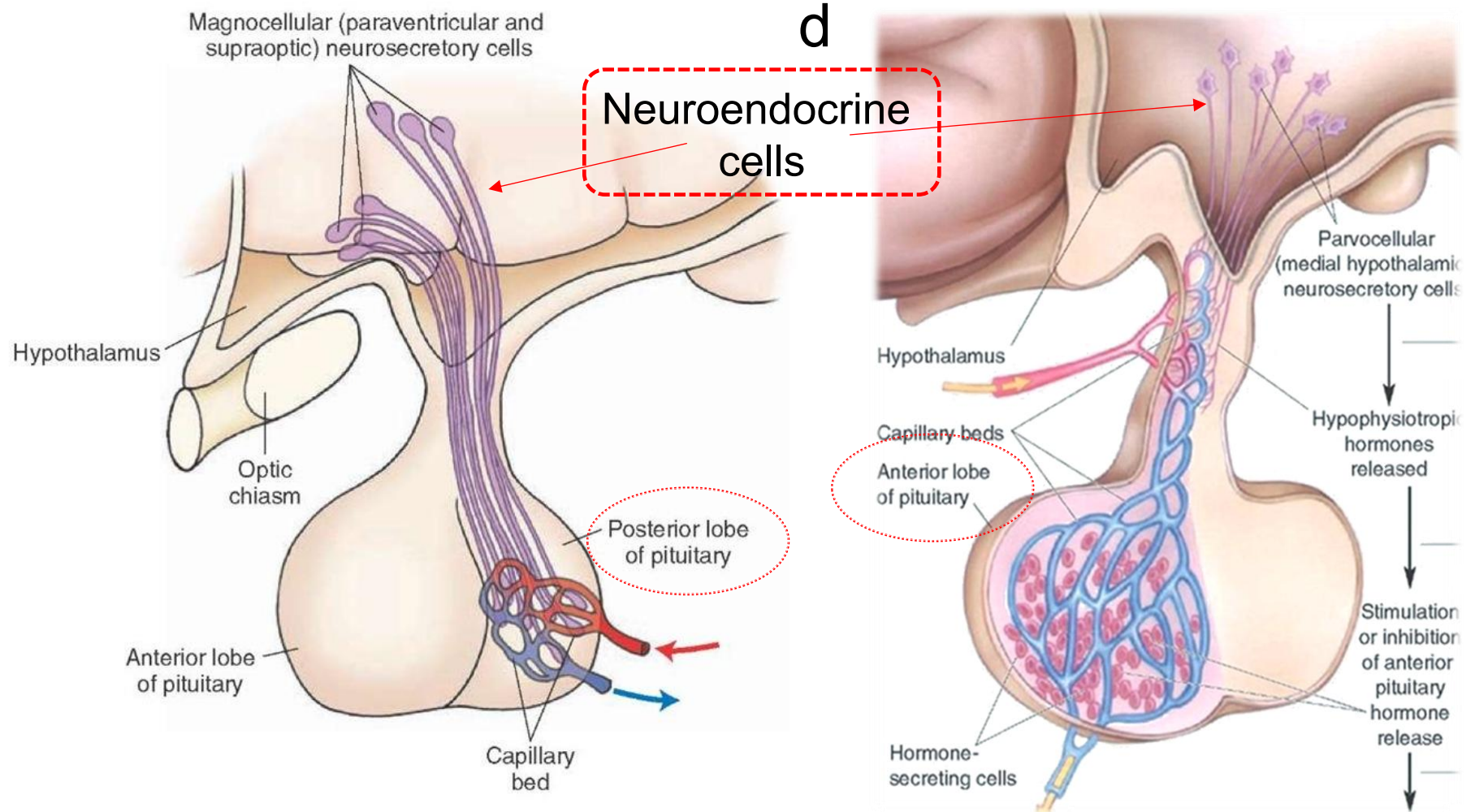
It has a close anatomical and functional relationship with the hypothalamus

2 lobes, each with distinct connection to the hypothalamus

adenohypophysis
pars distalis



Connections Between the Hypothalamus and Pituitary



• Magnocellular (large, long) neurons

- large cell bodies with long axons projecting down to (posterior) pituitary
- oxytocin and vasopressin (ADH)
- released into systemic circulation

• Parvocellular neurons (small, short)

- small cell bodies with shorter axons projecting to top of the pituitary stalk (infundibulum)
- released into hypophyseal portal circulation
- 6 hormones, referred to as the hypophysiotrophic hormones

Hormones of Hypothalamus

Hypophysiotrophic hormones

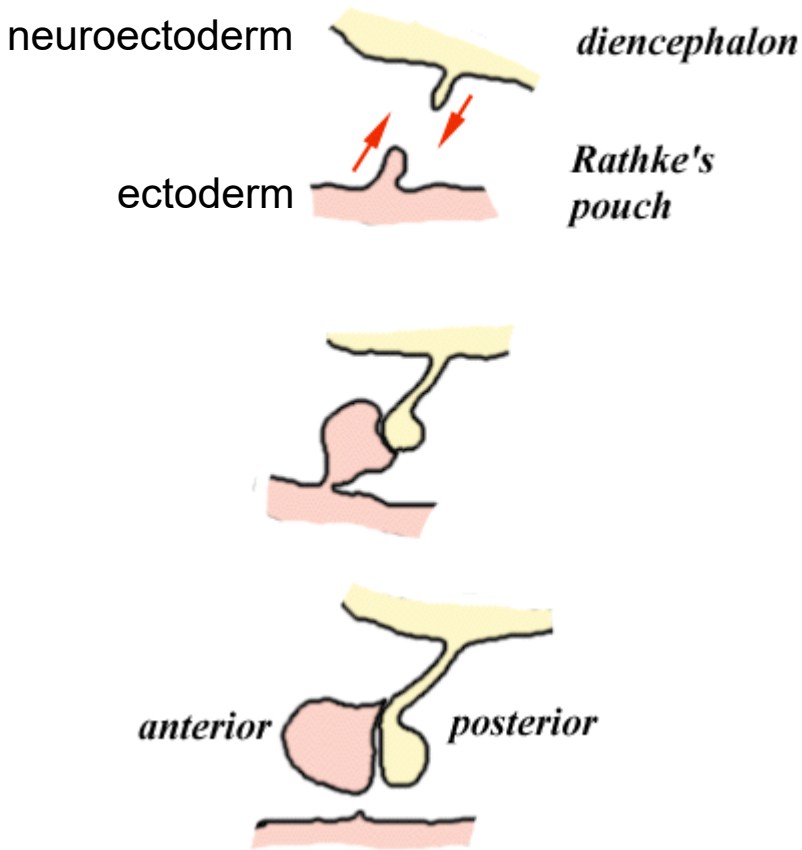
Hormone	Class	Function
Thyrotrophin Releasing Hormone (TRH)	Peptide (3 aa)	Stimulate TSH secretion from anterior pituitary
Corticotrophin Releasing Hormone (CRH)	Peptide (41 aa)	Stimulate ACTH secretion from anterior pituitary
Somatostatin (SST)	Peptide (14/28 aa)	Inhibits GH release from anterior pituitary
Growth Hormone releasing Hormone (GHRH)	Peptide (44 aa)	Stimulate GH secretion from anterior pituitary
Gonadotrophin Releasing Hormone (GnRH)	Peptide (10 aa)	Stimulate FSH and LH secretion from anterior pituitary
Dopamine	Catecholamine	Inhibits prolactin release from anterior pituitary
Vasopressin (AVP)	Peptide (9 aa)	Stimulates water reabsorption by kidneys
Oxytocin	Peptide (9 aa)	Stimulates uterine contractions during childbirth

Hormones of posterior pituitary

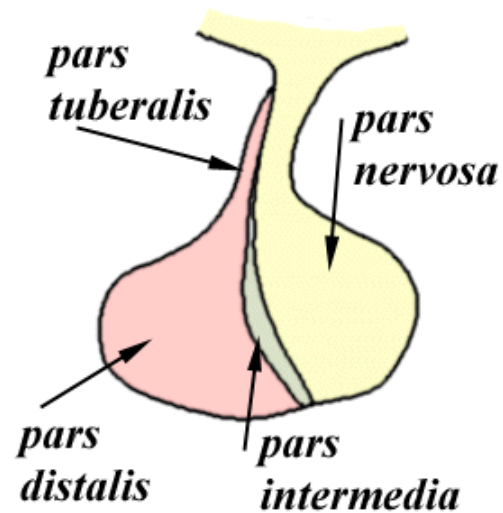
Posterior Pituitary

Pituitary Gland Development

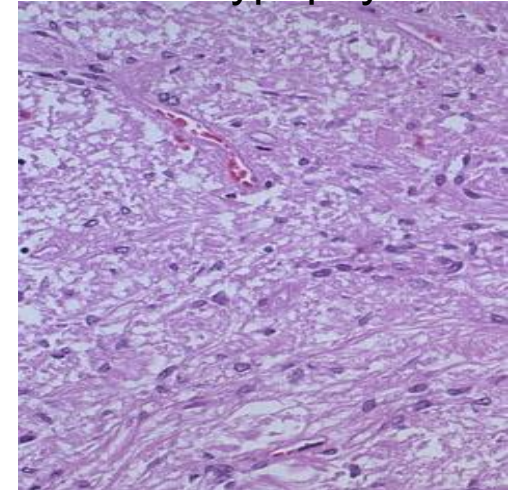
Pituitary Development



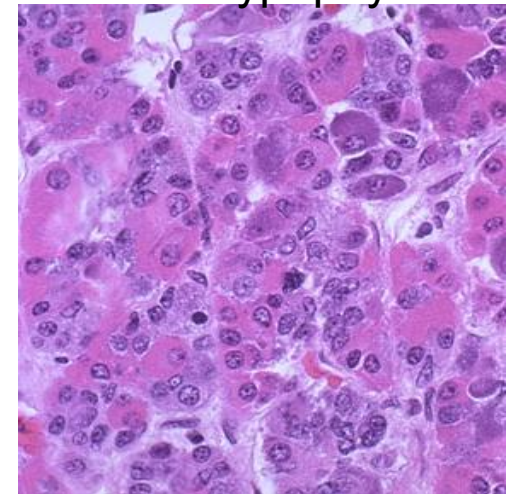
Pituitary Gland



neurohypophysis

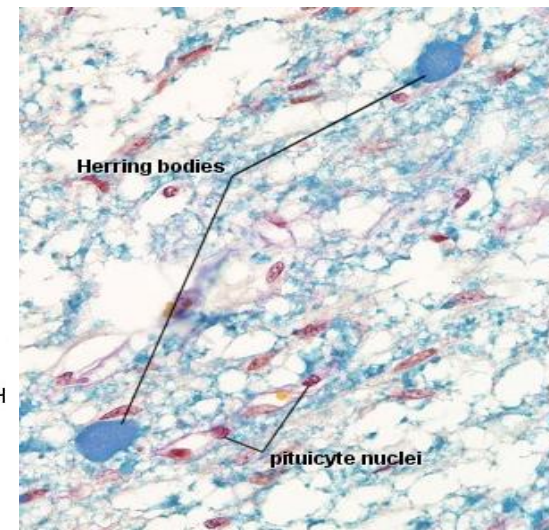
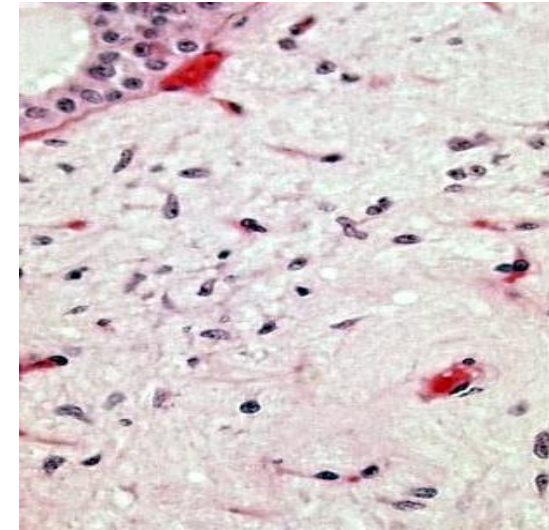
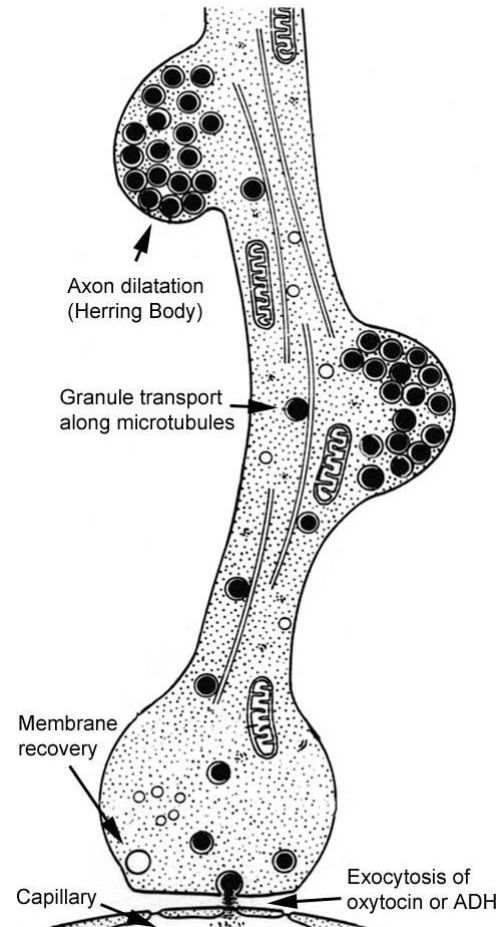


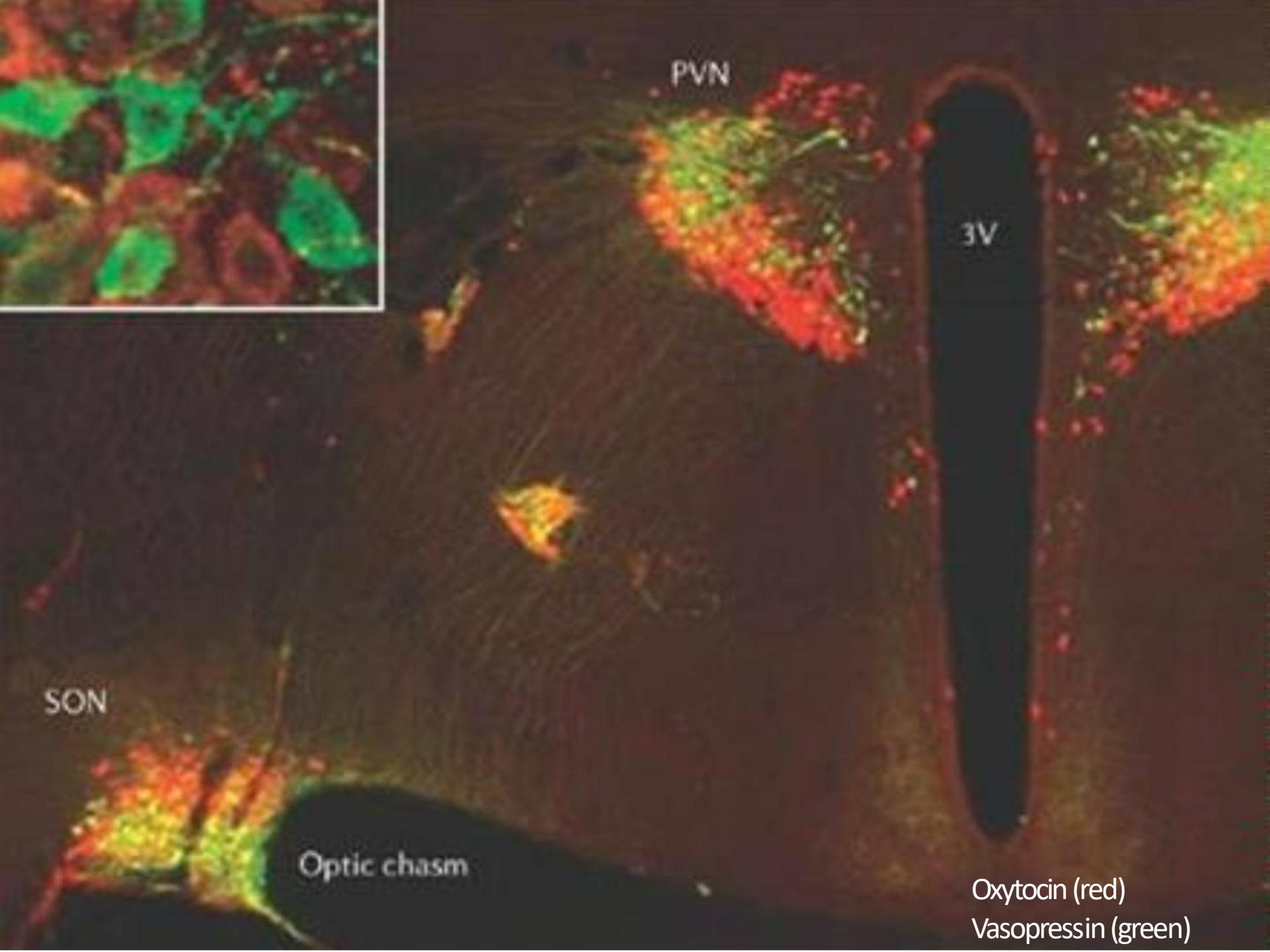
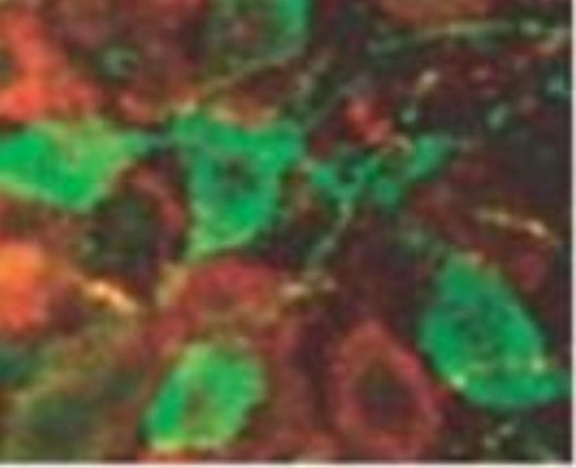
adenohypophysis



Functional Histology of Neurohypophysis

- Unmyelinated nerve axons of neurons cells whose cell bodies are located in hypothalamus
- Pituicytes (glial cells)
- Herring bodies
Secretory granules containing oxytocin and vasopressin in terminal region of axon





PVN

3V

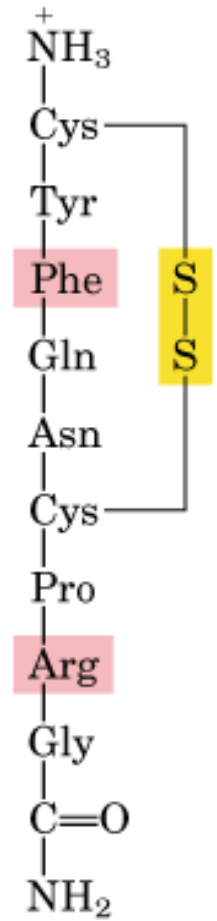
SON

Optic chiasm

Oxytocin (red)
Vasopressin (green)

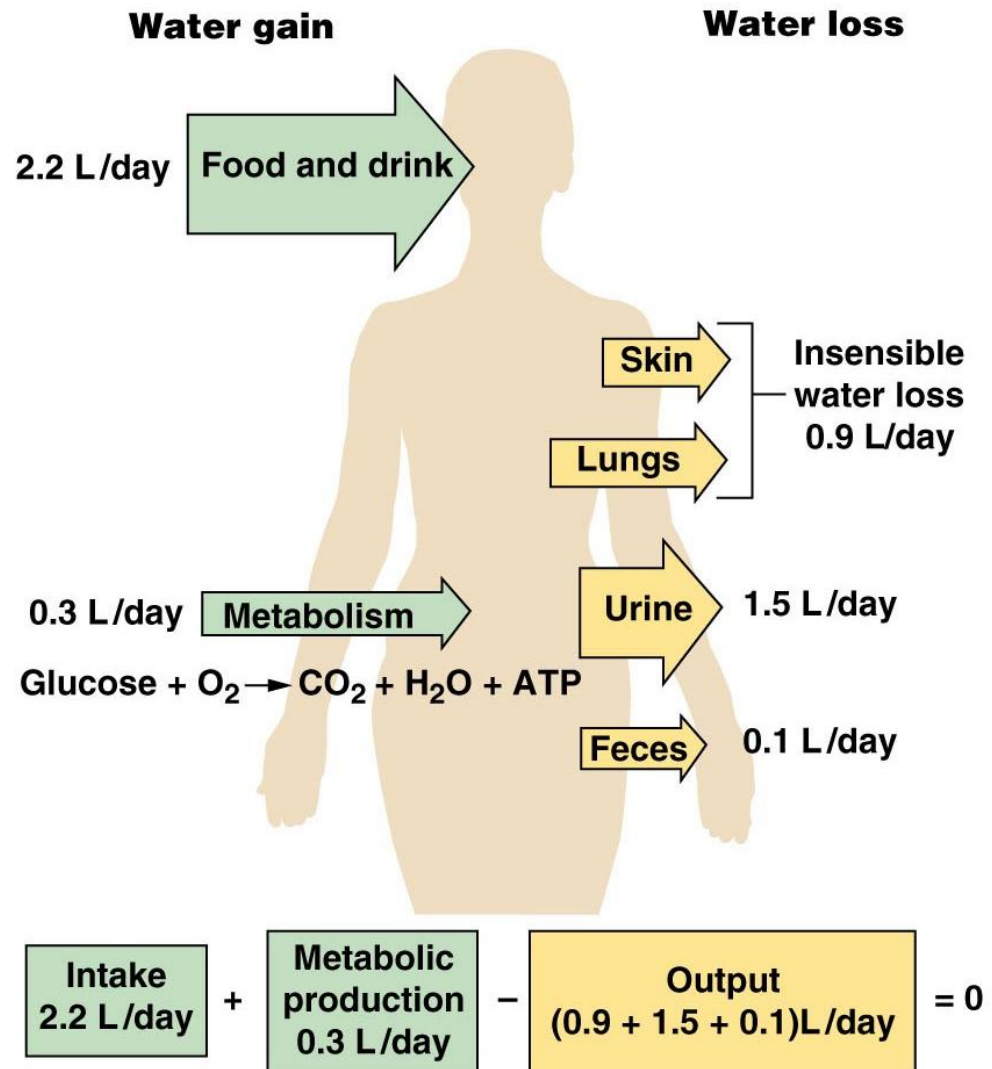
Vasopressin

- Also known as antidiuretic hormone (ADH)
- Acts through 2 different receptors
 - V_1 receptors abundant in vascular smooth muscle cells
 - activation of V_1 receptors leads to VSMC contraction ('vasopressin')
 - V_2 receptors abundant in renal tubule epithelial cells
 - activation of V_2 receptors leads to increased renal H_2O reabsorption
- Plays essential role in maintaining water balance



Water Balance in the Body

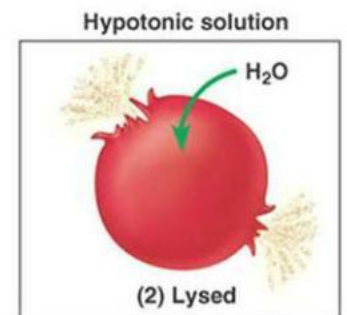
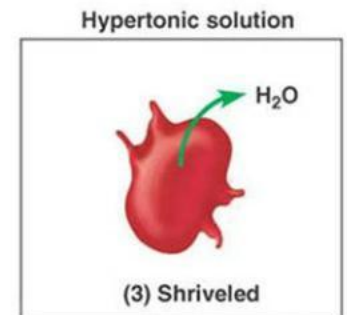
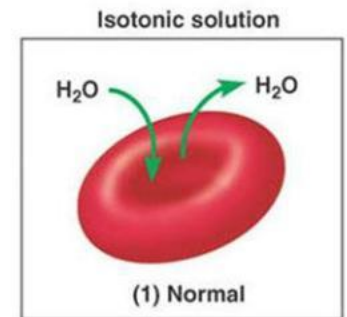
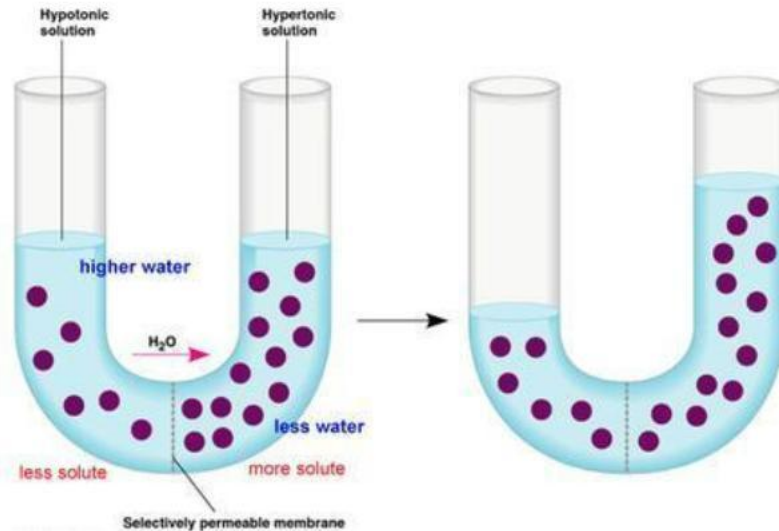
- H₂O makes up 50-60% of total body weight.
- Changes in H₂O content of the body affect osmolality of plasma
- This has important consequences for water movement into and out of cells/tissues, affecting cell volume and the conc. of many important cell and plasma solutes
- Controlling urinary output is a key parameter in maintaining water balance - AVP



Osmosis

Movement of **solvent** molecules (H_2O) across a selective permeability barrier (eg. cell membrane, endothelium)

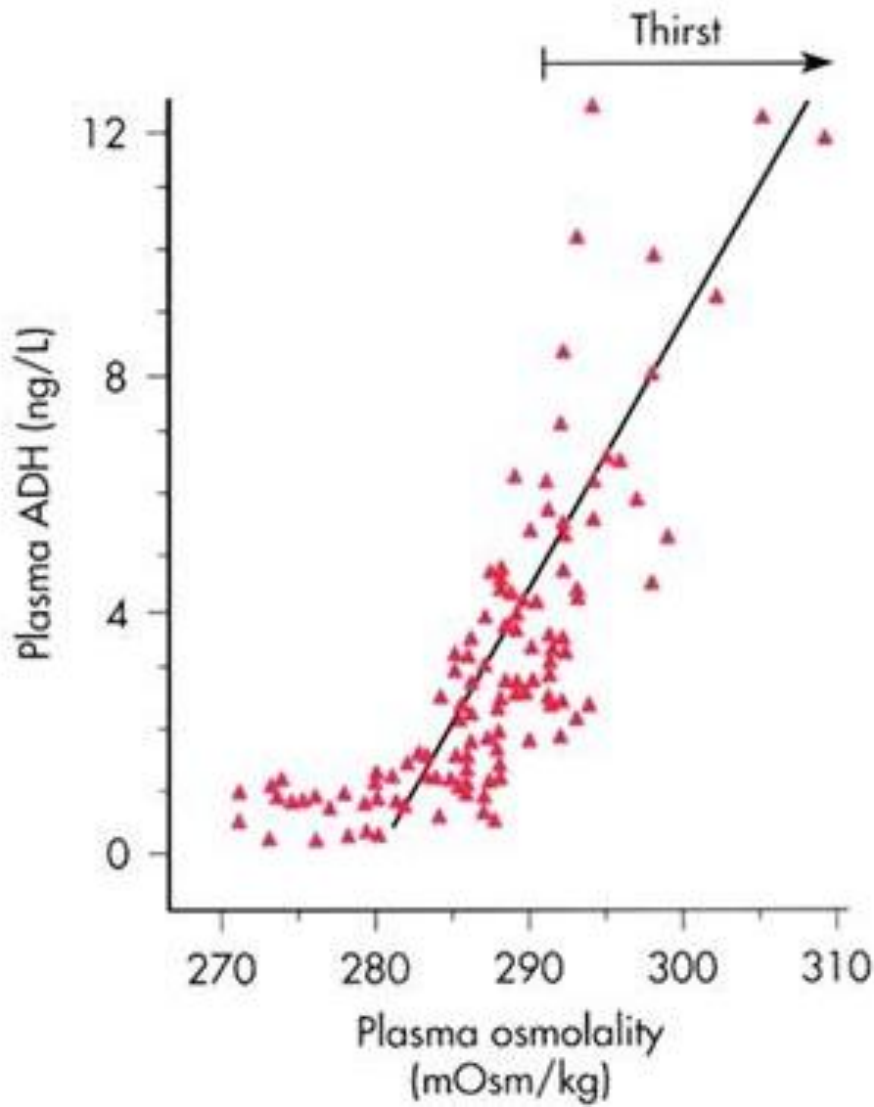
Body fluids (eg. blood, ECF) exert osmotic pressure (measured as osmolality)



The H_2O content of the body is the main determinant of osmolality
– regulation of body water content is known as **osmoregulation**

Osmolality is tightly controlled to avoid this

Plasma AVP and Osmolality



Plasma osmolality of $> 280\text{mOsm/kg}$ induces AVP release

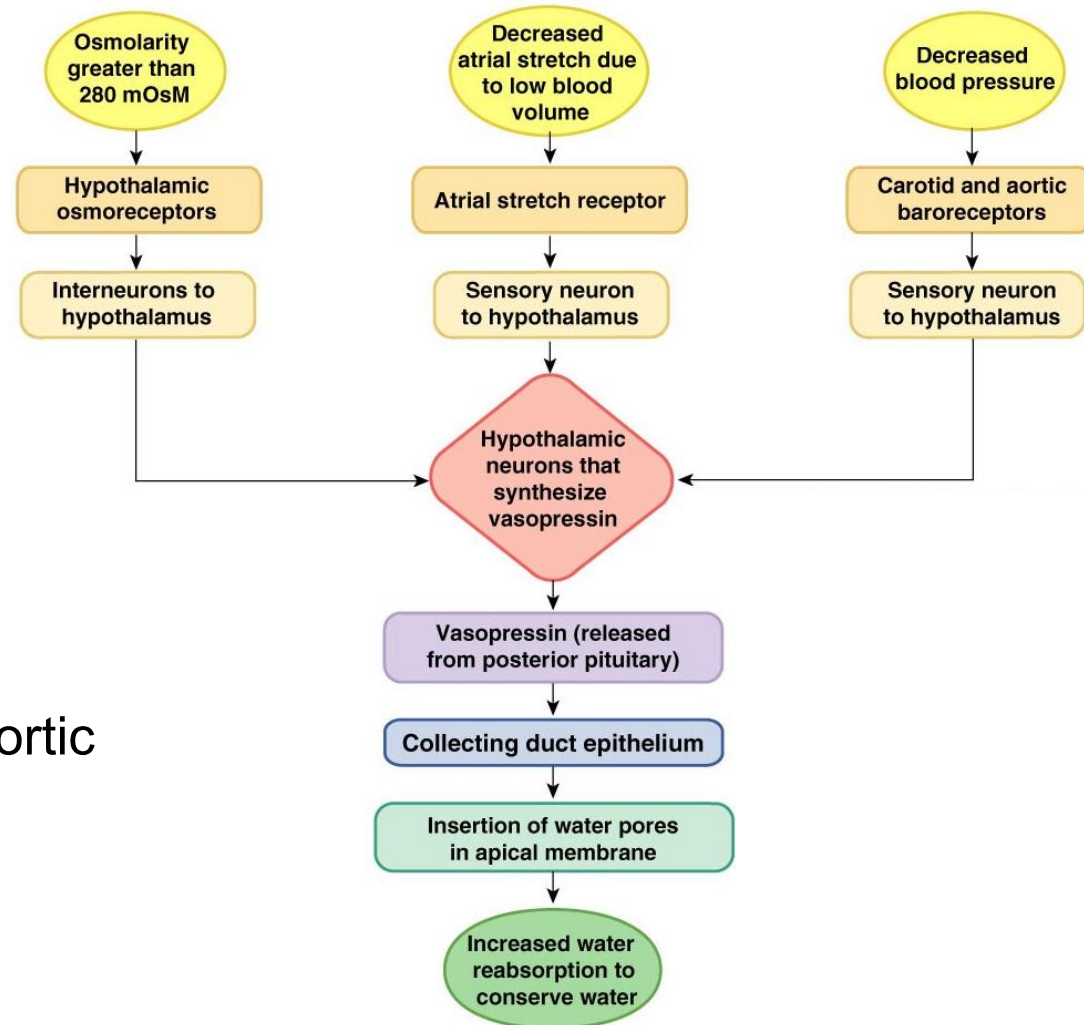
AVP acts on renal epithelial cells to promote reabsorption of H_2O

Plasma volume is increased and its osmolality decreased and urine volume is decreased and its osmolality is increased

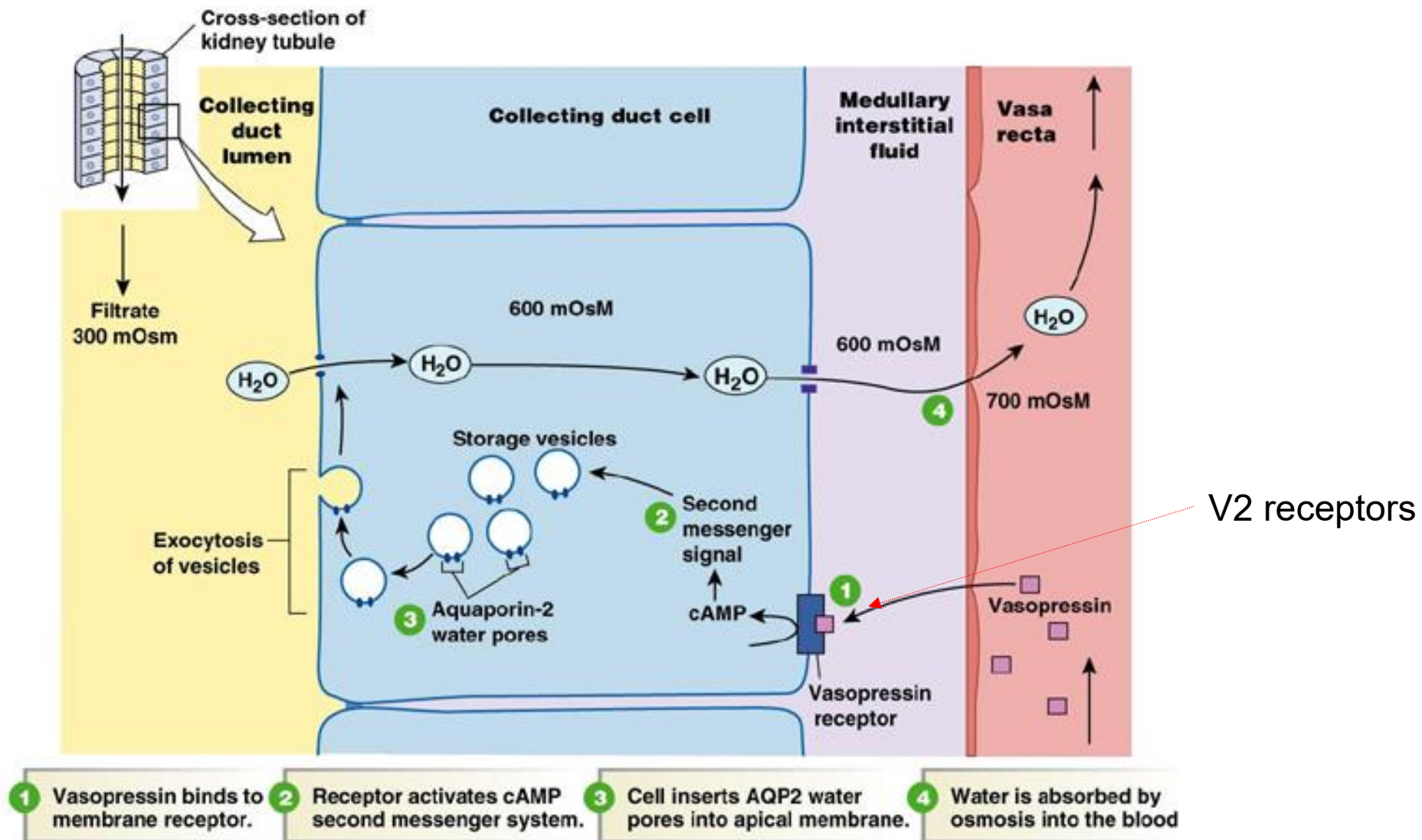
Independently, thirst centres in the hypothalamus are activated increasing the urge to consume H_2O

Factors Affecting AVP Release

- 3 stimuli control AVP release
- blood osmolality $> 280\text{mOsM}$ (sensed by hypothalamus osmoreceptors)
- hypovolemia ($<$ atrial stretch receptor activation)
- hypotension ($<$ carotid and aortic baroreceptor activation)



AVP Actions in Kidney Tubule Epithelia

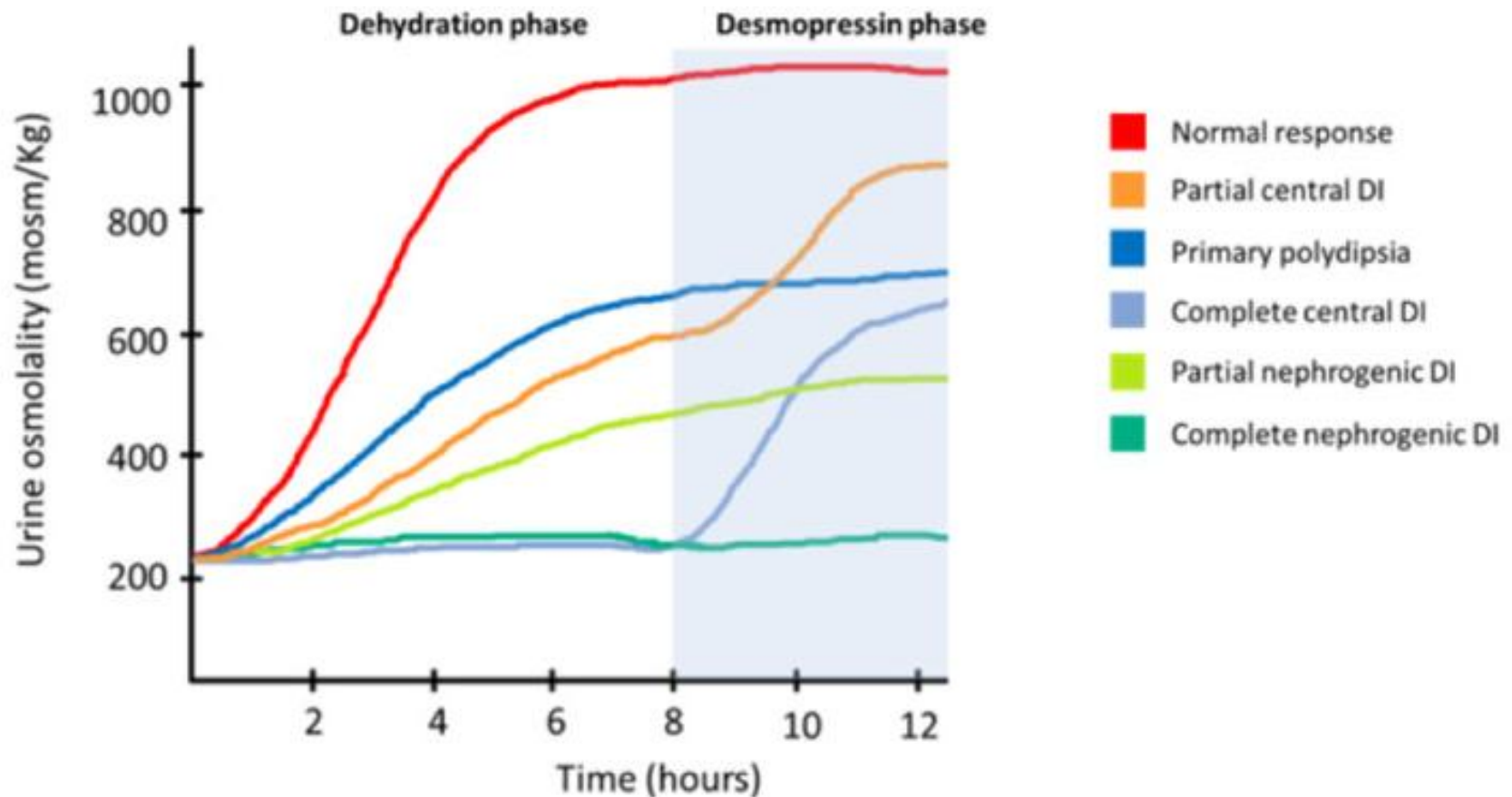


Diabetes Insipidus (DI) : AVP Deficiency or Resistance

- Pathophysiology
 - inability of kidney to reabsorb H_2O with excretion of large volumes of dilute urine (25l/day)
- Types
 - neural (cranial) DI – hypothalamus/pituitary lesion
 - nephrogenic DI – renal 'resistance' to AVP
 - inherited - mutations in V2 receptor (X linked) or aquaporin 2 (autosomal)
 - acquired - drug induced (amphotericin B)
- Diagnosis
 - water deprivation test
- Treatment
 - desmopressin (synthetic AVP)

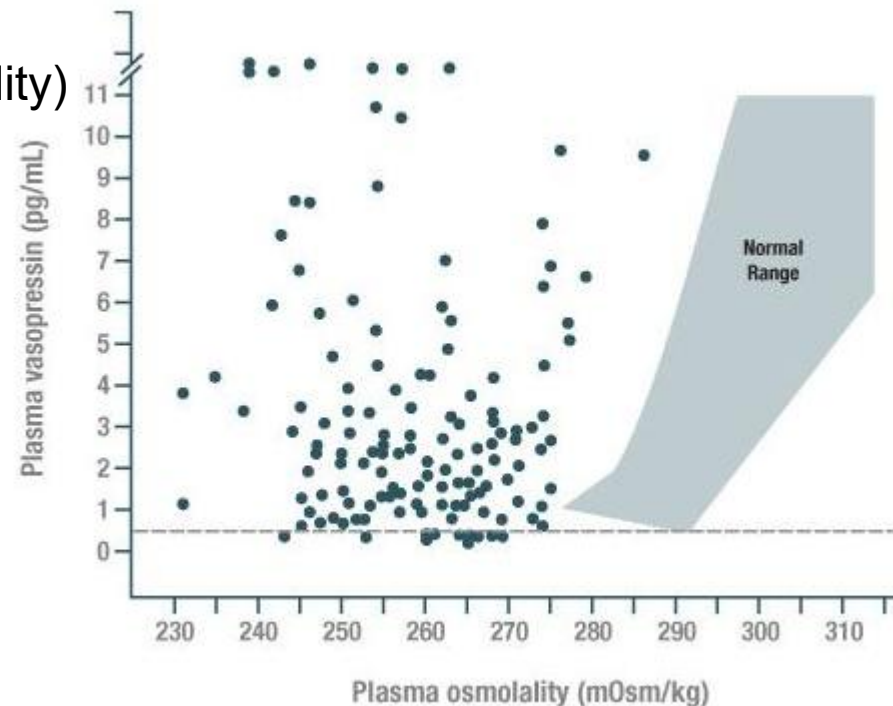
Water Deprivation Test

Deprive subjects of water for ~ 12 hrs and measure urine osmolality/volume
Administer desmopressin at 8hrs



Syndrome of Inappropriate Antidiuretic Hormone Secretion (SIADH)

- Inappropriate ADH secretion despite normal or increased plasma volume
 - $\uparrow$ H₂O reabsorption ($\downarrow$ plasma osmolality)
 - normo or hypervolemia (hypertension)
 - hyponatremia
 - low sodium and plasma osmolality can lead to acute edema of the brain
- Etiology
 - ectopic production of ADH by GIT and pulmonary tumours
- Water loading test
 - tests the kidney's ability to excrete a water load by suppressing ADH

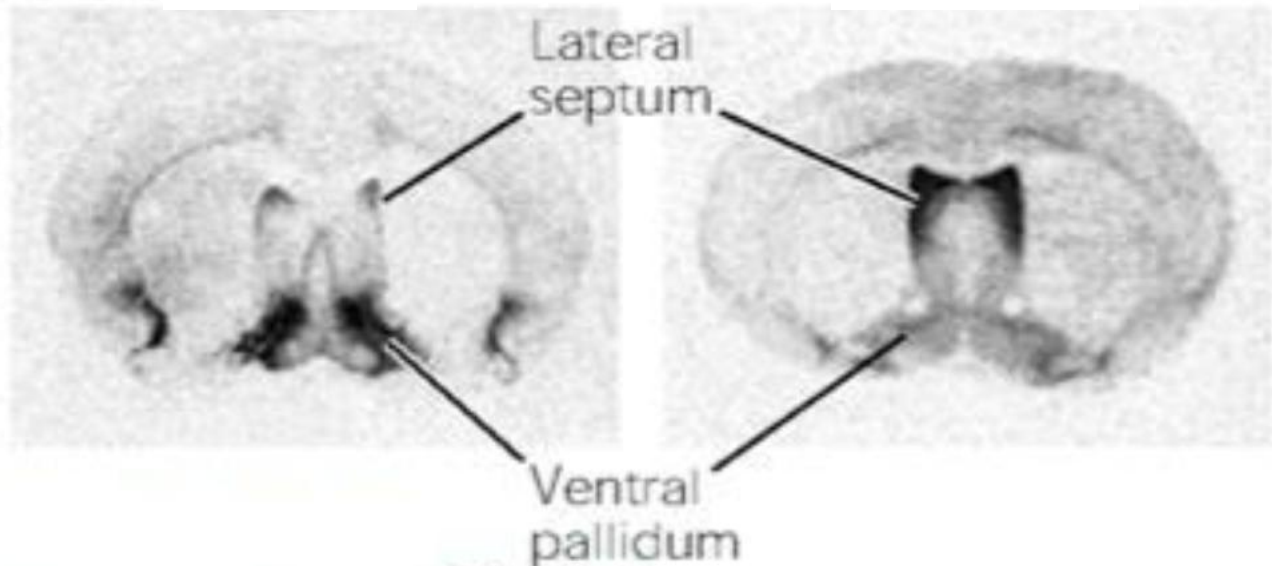


Patterns of AVP Receptor Expression in Brains of Meadow and Prairie Vole

Prairie

Meadow

Vasopressin
receptor
(V1aR)



Ventral pallidum is an area associated with trust, reward, motivation and bonding

Meadow vole = promiscuous

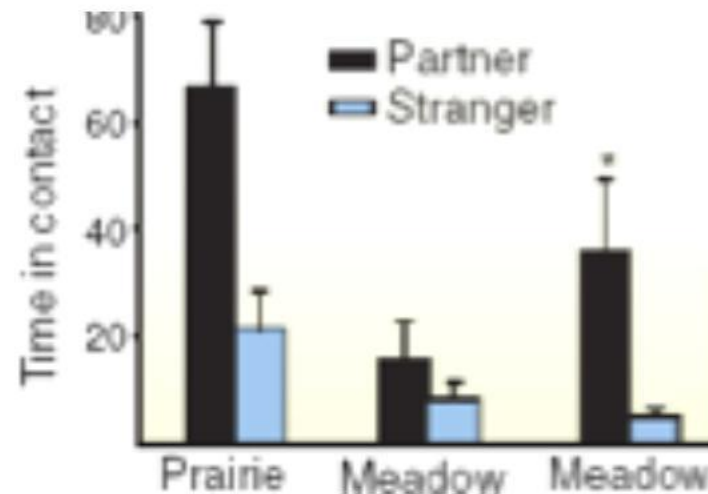
Prairie vole = faithful

Vasopressin – A Cure for Infidelity ?

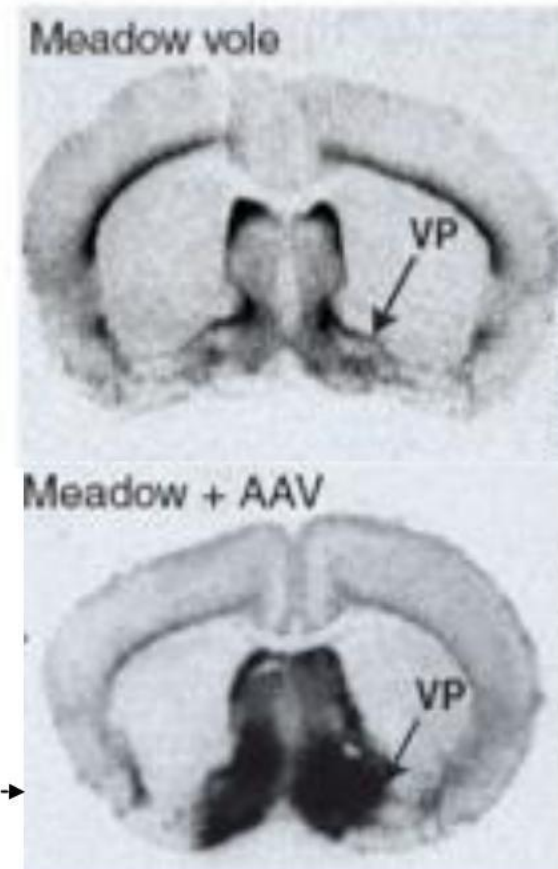
Enhanced partner preference in a promiscuous species by manipulating the expression of a single gene *Nature* 429, 754-757 (17 June 2004)

By transferring the vasopressin receptor gene to the pleasure centre of the naturally promiscuous male meadow vole, researchers were able to make it happily monogamous, a pattern of behaviour typical of its close relation, the prairie vole

Increased similarity to prairie vole behaviour

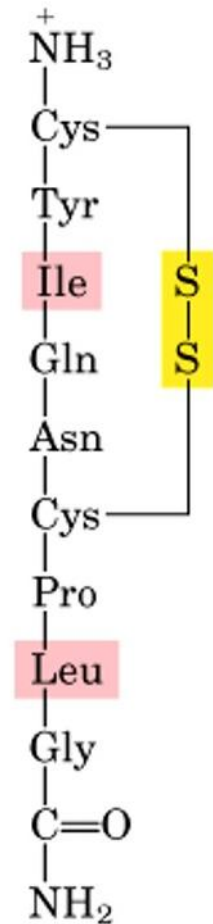


Increased AVP receptor expression in VP due to viral mediated gene transfer



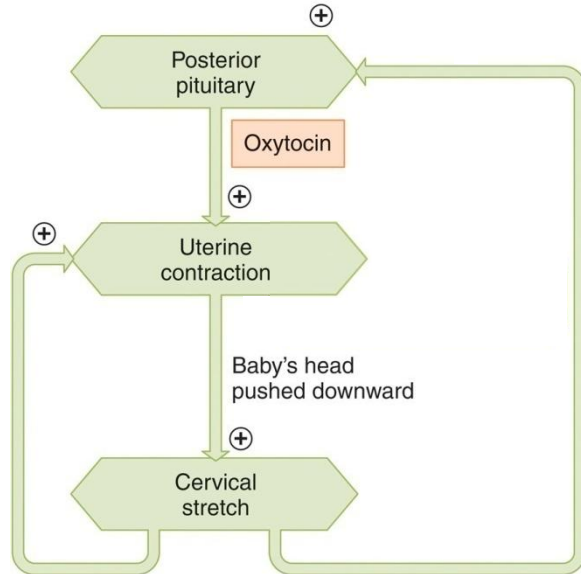
Oxytocin

- Structurally related to AVP
- Weak antidiuretic action (similarity to AVP)
- Physiological roles
 - parturition
 - lactation
 - social / mating / emotional behaviour

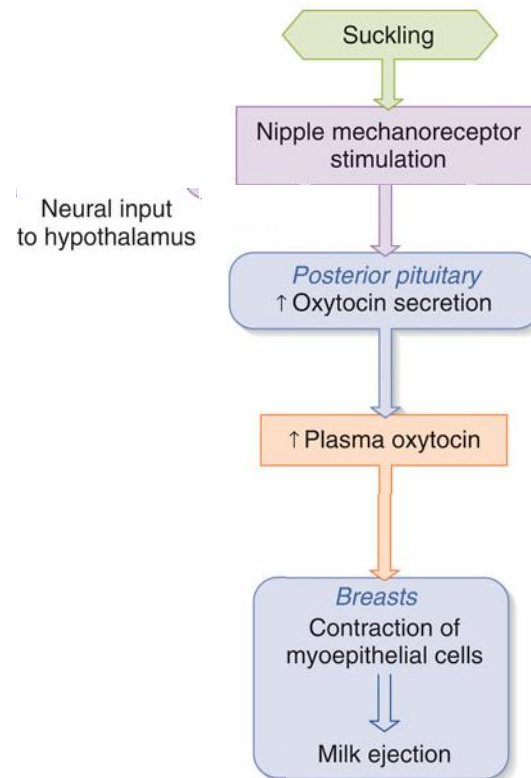


Physiological Role of Oxytocin

- Lactation
 - stimulates myoepithelial contractions to expel milk from mammary alveoli into ducts/collecting chamber ('milk let down reflex')



- Parturition
 - stimulates uterine contractions
 - can be used to induce labour



Oxytocin: The 'Cuddle Chemical'

- Post coital 'glow' – promotes partner bonding/monogamy
- Postpartum motherchild bonding – maternal nurturing behaviour
- Reduces fear and anxiety
- Promotes trust and empathy
- 'Scientists Probe Oxytocin Therapy for Social Deficits in Autism and Schizophrenia'
JAMA. 2011;305(7):659-661



The screenshot shows the Biotree website for 'Liquid Trust'. The header includes navigation links: HOME, ABOUT, PRODUCTS, and CONTACT. The main heading is 'BIOTREE'. Below it, there's a section 'OUR PRODUCTS' featuring 'Liquid Trust' as the 'World's first Oxytocin product, specially formulated to create a trusting atmosphere!'. A 'UK Supplier' badge and 'ORDER NOW!' button are present. Text asks if the user needs others to trust them more and provides a 'Read More?' link. A contact number '0845 3882498' is displayed. The 'Who is Using LiquidTrust?' section lists 'Salespeople', 'Managers & Employees', and 'Singles' with brief descriptions. The 'In The News' section features logos for WebMD, Scientific American, BBC News, and National Geographic, each with a 'More' button. A 'Sun' newspaper logo is also visible with the date '14/07/07'. The footer text reads 'Helping you with that all important first impression'.

HOME ABOUT PRODUCTS CONTACT

BIOTREE

ABOUT US

BioTree
Welcome to our new site. This site is currently being fully developed to bring you full details on our new exciting product, Verolab's Liquid Trust.

OUR PRODUCTS

Liquid Trust
World's first Oxytocin product, specially formulated to create a trusting atmosphere!

Do you need others to trust you more?
Could you sell more, love more and accomplish more if people trusted you more? [Read More?](#)

Contact Us for Our Product Range and Prices

Call Local Rate
0845 3882498

Who is Using LiquidTrust?

- Salespeople**
You can have the world at your fingertips when people trust you they want to buy from you...
- Managers & Employees**
If you have the confidence of the people you work with you will earn more and GET AHEAD FASTER
- Singles**
It's easy to meet new people when you have the power of Trust

In The News :
What the world's media is saying about Oxytocin

WebMD SCIENTIFIC AMERICAN BBC NEWS NATIONAL GEOGRAPHIC

More More More More

Exclusive UK News
Sun
14/07/07

Helping you with that all important *first impression*

Friday 18th Sept

‘Anterior Pituitary Physiology’ @ 12:00 in C005

Biochemistry Practical

15:00 – 17:00 in Science Centre East (E2.53 and 2.54)

Only attend the session you are assigned to – consult the listings on Brightspace (Laboratory Practical –GTT folder)

Bring lab coat, graph paper and writing materials

Print out and partially fill in a lab report before the practical (title, background, aims, methods) – leave some blank space to enter your results you will complete it on the day

Please read the supplementary material in the Brightspace folder

For those wishing to do their own personal GTT, please print out and sign the consent form and bring it to the practical